

		voltmeter to decrease.
20	C	Let resistance of variable resistor be R, p.d. across lamp be V_{lamp}. Option A: Incorrect $R = 0 \, \Omega$, $V_{\text{lamp}} = E$

Qn	Answer	Solution
		$R = \max$, $V_{\text{lamp}} = V_{\text{fixed resistor}}$ Option B: Incorrect $R = 0 \, \Omega$, $V_{\text{lamp}} = 0 \, \text{V}$ $R = \max$, $V_{\text{lamp}} = V_{\text{variable resistor}}$ Option C: Correct $R = 0 \, \Omega$, $V_{\text{lamp}} = 0 \, \text{V}$ $R = \max$, $V_{\text{lamp}} = E$ Option D: Incorrect $R = 0 \, \Omega$, $V_{\text{lamp}} = E$ $R = \max$, $V_{\text{lamp}} = E - V_{\text{variable resistor}}$
21	B	In order for the voltmeter to have a reading of 12 V, the resultant resistance